Murali Sai Buddakkagari Venkata
AI Engineer

LLM Agents, RAG & Production AI Systems
+1 (201)-275-6592 | murali140824@gmail.com | LinkedIn | GitHub | Portfolio

EXPERIENCE

Dreamline AI – AI Engineer **Sep 2026 – Present**
Community Dreams Foundation · Remote · Data layer (PostgreSQL, Supabase)
- Designed and secured a Supabase/PostgreSQL schema extension (Row Level Security, audit triggers) that lets a 141-program catalog pre-fill applications, passing 7/7 acceptance criteria locally with 0 of 141 records altered.
- Designed the requirement layer: a 21-key registry, a three-valued JSON condition engine, and a readiness API, packaged as an idempotent, rollback-ready release.
- Adapted the migration to the live schema in a Lovable planning pass (reused existing columns, separated eligibility from requirements, fixed an audit-trigger blocker) after baselining production data; SQL submitted for review.

Querkey Inc – Data Science Intern **Jun 2025 – May 2026**
CA, US · Multimodal AI & MLOps (Gemini, RAG, MLflow)
- Cut production ML workflow failures 60% across 100K+ records with a Dockerized observability stack (MLflow, FastAPI, ChromaDB, HuggingFace monitoring) enabling real-time failure detection.
- Worked directly with healthcare stakeholders to build and deploy a proof-of-concept real-time Patient Flow Tracking System using multimodal AI (Gemini 2.5 Flash) and RAG to structure patient notes for insight retrieval.
- Authored a 30-query SQL bank to ground an AI chatbot's RAG answers over hospital data, and built a Superset REST API integration for on-demand dashboard charts.

View Group India – AI and Automation Intern **Jan 2024 – Aug 2024**
Formerly View Synergy · KA, IND · Full-stack (React, .NET, MySQL)
- On a cross-functional team modernizing a live platform, reviewed React, .NET Azure Functions, and MySQL code, root-causing 2 recurring exceptions and surfacing SQL-injection vulnerabilities in a 94-issue security assessment.
- Built and tested a new Roles/Designations CRUD feature end to end (MySQL stored procedures, .NET, React).
- Audited Redis usage across 6 backend microservices and documented Microsoft Entra ID app registration and role setup, supporting a proposal to replace a custom JWT/BCrypt authentication system.

PROJECTS

SEC EDGAR RAG Filing Analyzer
- Architected and deployed a production RAG system over 8,232 SEC 10-K chunks across 5 companies (FastAPI, ChromaDB, BM25 plus dense retrieval with Reciprocal Rank Fusion, cross-encoder reranking, RBAC information barriers) on Cloud Run, verified by a 504-test CI suite and a 54-question evaluation (0.853 faithfulness, 0.936 citation accuracy, 0.981 refusal correctness).
- Designed the agent harness and context engineering (entity detection pinning retrieval to the named filing, per-entity budgets, bounded agentic retry, prompt-injection and MNPI guardrails) and exposed it as an MCP server for Claude Desktop and Cursor with RBAC and guardrails preserved.

Agent Orchestration System with Tool Use, Memory, and Human in the Loop
- Architected and shipped a production multi-agent system (19 REST endpoints) on a 3-layer LangGraph hierarchy (planning supervisor, parallel specialists across dependency frontiers, reflective reviewer with a rework loop), multi-model routing and two-tier Redis/pgvector memory, validated by 106/113 tests on a live deployment.
- Built a custom MCP server (6 agent skills), OpenTelemetry tracing with per-task cost tracking, and a Human-in-the-Loop system (4 escalation severity levels, audit logging), using spec-driven development with Claude Code.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript (React), C# (.NET)
AI & Machine Learning: Multi-Agent Orchestration, Agent Harness & Context Engineering, Prompt Engineering, MCP & Agent Skills, LangGraph, LangChain, Retrieval-Augmented Generation, Reciprocal Rank Fusion, Agent Evaluation (RAGAS), Multimodal AI, Embeddings, Vector Databases, PyTorch
Infrastructure & DevOps: FastAPI, ChromaDB, Docker, OpenTelemetry (Observability), Google Cloud Platform, Azure, PostgreSQL, GitHub Actions, CI/CD, Git, REST APIs, OpenAI/Claude/Gemini APIs

EDUCATION

Stevens Institute of Technology **May 2026**
Master of Science – Applied Artificial Intelligence *NJ, US*

PES University **May 2024**
Bachelor of Technology – Computer Science (Specialization in ML) *KA, IND*

PUBLICATIONS AND CERTIFICATIONS

- Adaptive Ensemble Detection for Medical Imaging, M.S. Capstone, Stevens (Advised by Prof. Kevin Lu): size-conditioned Weighted Box Fusion of Faster R-CNN, Mask R-CNN, and YOLO26 reaching 0.71 mAP@0.5 on the 2018 Data Science Bowl.
- Best Research Paper Award, ICICC 2024: WorkFit, a real-time posture monitoring system using CNN-based vision and flex sensor fusion.